

ESTIMATION OF BINOMIAL PROPORTIONS FROM POOLED SAMPLES USING AN OBJECTIVE PRIOR

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Summary: Pooled testing is where units are pooled together and tested as a group rather than individually. Pooled testing has been used in many fields of study, as individual testing can be too time consuming and pooled testing is more cost-effective. In this paper we will look into confidence intervals for linear functions of binomial proportions from pooled samples. We will investigate the performance of credibility (Bayesian confidence) intervals for a single proportion as well as the difference of two binomial proportions estimated from pooled samples. An objective (non-informative) prior, the Jeffreys prior, will be used. Results from the Bayesian method will be compared to results from some known classical methods. These intervals will be compared with each other in terms of coverage, left non-coverage, right non-coverage, symmetry and interval length.

1. Introduction

Pooled testing was first introduced by Dorfman (1943) for medical screening purposes to identify infected individuals. In the case of pooled testing units are pooled together and tested as a group rather than individually. Pooled testing has been used in many fields of study since individual testing can be too time consuming and pooled testing is more cost effective. Extensive arguments for cost effectiveness of pooling are given in Gastwirth and Johnson (1994) and Johnson and Gastwirth (2000).

Chiang and Reeves (1962) and later Sobel and Elashoff (1975) derived maximum likelihood estimators for equal pool sizes. The theoretical properties of these estimators were investigated by Tu, Litvak and Pagano (1995). Walter, Hildreth and Beaty (1980) extended maximum likelihood methods to unequal pool sizes and applied a simple Newton-Raphson algorithm to estimate infection prevalence of yellow fever virus in mosquitoes. Hepworth (1996) considered the sequential testing

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of pools by constructing exact confidence intervals for problems involving unequal pool sizes. The exact methods were however only feasible for less than four pool sizes. Hepworth (2005) also considered asymptotic interval estimation methods for pools of different sizes. He investigated four methods, two based on the distribution of the maximum likelihood estimator (MLE), one on the score statistic and one on the likelihood ratio statistic. He recommended the method based on the score statistic with a correction for skewness.

As mentioned by Biggerstaff (2008) comparison of proportions estimated from pooled samples has received little attention, even in the case of equal pool sizes. Walter et al. (1980) calculated the Wald confidence interval for the difference between two proportions in the case of unequal pool sizes. McCann and Tebbs (2007) on the other hand used a multiple testing procedure for pairwise differences of proportions if the pool sizes are all equal. Biggerstaff (2008) derived asymptotic confidence intervals for the difference between two proportions in the case of equal as well as unequal pool sizes. He presented seven confidence interval methods and recommended the skewness-corrected profile score interval. In the next sections the point estimates and confidence intervals calculated by Biggerstaff (2008) will be compared to our Bayesian methods.

From a Bayesian point of view not much research has been done on pooled testing. Chick (1996) presented Bayesian models for obtaining posterior distributions of the unknown proportion p . He used the $Beta(a, b)$ prior with different values of a and b to calculate point estimates and equal-tailed credibility intervals. The methods were applied on pooled test data for gene transfer experiments and limiting dilution assay data for immunocompetency studies. Hanson, Johnson and Gastwirth (2006) on the other hand considered the problem of screening HIV among blood donors in the United States. They used a two-stage sampling procedure and developed a Bayesian method that allows for sampling multiple sites in a specific region. The reason for first-stage pooling was for cost effectiveness and the development of a “gold standard” test. As in Gastwirth and Johnson (1994) independent beta priors were used. The importance of using good prior densities were evident from the examples.

Instead of using a binomial pooled sampling approach as in Biggerstaff (2008), Pritchard and Tebbs (2011) argued that negative binomial pooled sampling might be preferred when the prevalence p is known to be small. For the special case $M = 1$ (number of distinct pool sizes) and common pool size $m > 1$, they used a Bayesian procedure to develop point estimates and credibility intervals for p . As prior they used the Jeffreys prior for the negative binomial distribution which is, according to them, appropriate where there is a lack of knowledge about the parameter p . Although the Jeffreys prior is improper, the posterior distribution is shown to be proper. The Bayesian procedure was applied to pooled mosquito testing for West Nile Virus.

In this paper point estimates and credibility (Bayesian confidence) intervals will be calculated for a single proportion as well as the difference between two binomial proportions from pooled samples with unequal pool sizes. The Jeffreys prior will be used for the Bayesian method, and equal-tailed as well as highest posterior density (HPD) intervals will be considered. In Section 2 theoretical aspects of the Jeffreys prior and the posterior distribution will be discussed. Two simulation studies will be considered in Section 3, and in Section 4 the Bayesian procedure on the estimation of the infection prevalence of yellow fever virus in mosquitoes will be illustrated. For the simulation studies and application, the Bayesian calculations will be compared to the results obtained by Biggerstaff (2008). The discussion and conclusion will be given in Section 5.

2. Prior and Posterior Distributions for Binomial Proportions from Pooled Samples

Assume that the proportion of successes in a given population is p . We will refer to an infected individual as a success in a binomial trial. Using the notation from Biggerstaff (2008), let N individuals be sampled independently from the population, and then be grouped into pools. The size of a pool will be indicated by m_i , for $i = 1, 2, \dots, M$, where M is the number of distinct pool sizes, let n_i be the number of pools of size m_i , and let X_i be the number of the n_i pools that are positive. Assume that $X_1, X_2, \dots, X_M$ are independent binomial random variables with $X_i \sim \text{Bin}(n_i, 1 - (1 - p)^{m_i})$.

The likelihood function is given by

$$L(p | x_1, x_2, \dots, x_M) \propto \prod_{i=1}^M \left\{ [1 - (1 - p)^{m_i}]^{x_i} [(1 - p)^{m_i}]^{n_i - x_i} \right\}. \quad (1)$$

2.1. The Prior

The Fisher information was derived by Walter et al. (1980), and is given by

$$I(p) = \sum_{i=1}^M \left\{ \frac{m_i^2 n_i (1 - p)^{m_i - 2}}{[1 - (1 - p)^{m_i}]} \right\}.$$

The Jeffreys prior, from Jeffreys (1939), is proportional to the square root of the determinant of the Fisher information and is given by

$$\begin{aligned} \pi(p) &\propto |I(p)|^{\frac{1}{2}} \\ \therefore \pi(p) &\propto \left(\sum_{i=1}^M \left\{ \frac{m_i^2 n_i (1 - p)^{m_i - 2}}{[1 - (1 - p)^{m_i}]} \right\} \right)^{\frac{1}{2}}. \end{aligned} \quad (2)$$

In Figure 1 graphs of the following Jeffreys priors are given.

1. $\pi_1(p) \propto \left\{ \frac{(1-p)^{48}}{[1-(1-p)^{50}]} \right\}^{\frac{1}{2}}$, i.e. $M = 1$, $m = 50$ and $n = 20$;
2. $\pi_2(p) \propto \left\{ \frac{2500(1-p)^3}{[1-(1-p)^5]} + \frac{5000(1-p)^8}{[1-(1-p)^{10}]} \right\}^{\frac{1}{2}}$, i.e. $M = 2$, $m_1 = 5$, $m_2 = 10$, $n_1 = 100$ and $n_2 = 50$;
3. $\pi_3(p) \propto \left\{ \frac{2000(1-p)^8}{[1-(1-p)^{10}]} + \frac{5000(1-p)^{23}}{[1-(1-p)^{25}]} + \frac{30000(1-p)^{48}}{[1-(1-p)^{50}]} \right\}^{\frac{1}{2}}$, i.e. $M = 3$, $m_1 = 10$, $m_2 = 25$, $m_3 = 50$, $n_1 = 20$, $n_2 = 8$ and $n_3 = 12$;
4. $\pi_4(p) \propto \left\{ \frac{500(1-p)^3}{[1-(1-p)^5]} + \frac{4000(1-p)^8}{[1-(1-p)^{10}]} + \frac{7500(1-p)^{23}}{[1-(1-p)^{25}]} + \frac{10000(1-p)^{48}}{[1-(1-p)^{50}]} \right\}^{\frac{1}{2}}$, i.e. $M = 4$, $m_1 = 5$, $m_2 = 10$, $m_3 = 25$, $m_4 = 50$, $n_1 = 20$, $n_2 = 40$, $n_3 = 12$ and $n_4 = 4$.

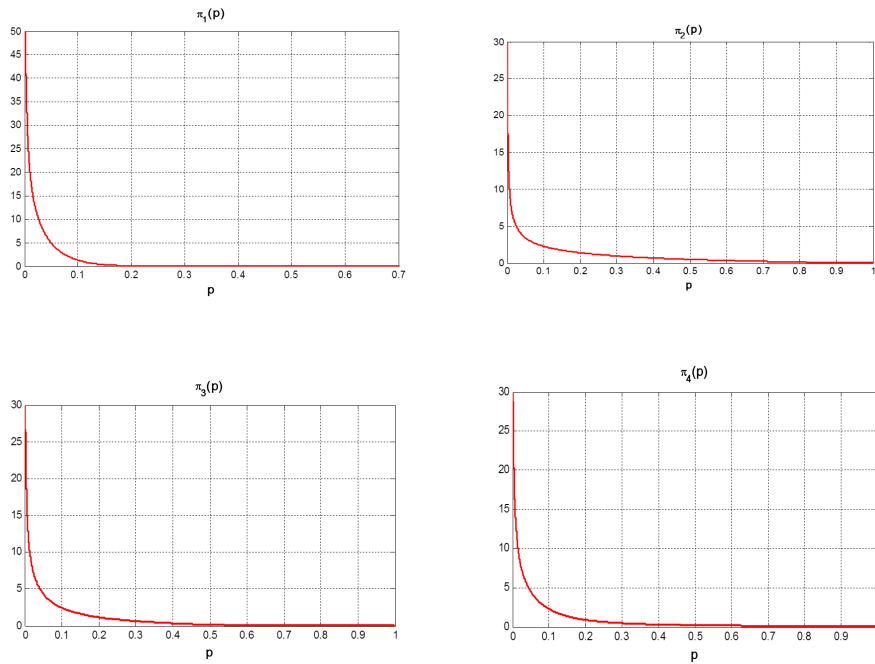


Figure 1: Illustration of the prior distributions.

From the graphs in Figure 1 it can be seen that the Jeffreys prior allocates much more mass to small values of p than to large values, which seems to be appropriate in this context.

The Jeffreys prior has some good properties:

- (i) It is defined for most distributions.
- (ii) It is invariant under one-to-one reparameterisation of parameters.
- (iii) It is proper for the binomial parameter p , but improper for the negative binomial distribution. However, in most cases it will yield proper posterior distributions.
- (iv) In the absence of nuisance parameters, the Jeffreys prior is a first-order probability matching prior, which means that the Jeffreys prior interval although Bayesianly constructed has good frequentist properties. According to Brown, Cai and DasGupta (2001) the Jeffreys prior interval is a continuity corrected version of the Clopper-Pearson interval.
- (v) It arises from many other approaches, such as minimum description length and various entropy approaches.
- (vi) It is therefore a clear winner in the absence of nuisance parameters.

A weakness of the Jeffreys prior is that it depends on the statistical model. The Jeffreys priors for the binomial and negative binomial distributions are not the same for the parameter p . It is also not always the best prior for higher-dimensional parameters. For further details about the Jeffreys prior, see Brown et al. (2001), Berger (2009) and Ghosh (2011).

2.2. The Posterior Distribution

From Equations 1 and 2 it follows that the posterior distribution is given by

$$\begin{aligned}\pi(p|data) &\propto \pi(p) \times L(p|data) \\ &\propto \left(\sum_{i=1}^M \left\{ \frac{m_i^2 n_i (1-p)^{m_i-2}}{[1-(1-p)^{m_i}]} \right\} \right)^{\frac{1}{2}} \\ &\quad \times \prod_{i=1}^M \left\{ [1-(1-p)^{m_i}]^{x_i} [(1-p)^{m_i}]^{n_i-x_i} \right\} \quad \text{for } 0 \leq p \leq 1. \quad (3)\end{aligned}$$

If $M = 1$, $m_1 = m$, $n_1 = n$ and $x_1 = x$, it follows from Equation 2 that

$$\begin{aligned}\pi(p) &\propto \left\{ \frac{m^2 n (1-p)^{m-2}}{[1-(1-p)^m]} \right\}^{\frac{1}{2}} \\ &\propto [(1-p)^m]^{\frac{1}{2}-\frac{1}{m}} [1-(1-p)^m]^{-\frac{1}{2}}, \quad (4)\end{aligned}$$

and the posterior distribution for this prior is therefore given by

$$\pi(p|data) = \frac{m}{B\left(x+\frac{1}{2}, n-x+\frac{1}{2}\right)} [(1-p)^m]^{n-x+\frac{1}{2}-\frac{1}{m}} [1-(1-p)^m]^{x-\frac{1}{2}}. \quad (5)$$

From Equation 5 we can see that the posterior distribution is proper when $M = 1$ (equal pool sizes). It is imperative that the posterior distribution is proper for all values of M . We therefore prove in the following theorem that the posterior distribution given in Equation 3 is proper. This means that it is integrable and therefore a well-defined posterior distribution.

Theorem 1. The posterior distribution

$$\pi(p|data) \propto \left(\sum_{i=1}^M \left\{ \frac{m_i^2 n_i (1-p)^{m_i-2}}{[1-(1-p)^{m_i}]} \right\} \right)^{\frac{1}{2}} \prod_{i=1}^M \left\{ [1-(1-p)^{m_i}]^{x_i} [(1-p)^{m_i}]^{n_i-x_i} \right\}$$

is proper for a M , $x_i \in \{0, 1, \dots, n_i\}$ and $0 < p \leq 1$.

Proof. We need to divide this proof into two parts. Firstly we consider the case if at least one $x_i > 0$ and secondly we consider the case where all $x_i = 0$.

Let at least one $x_i > 0$, say $x_1 > 0$, then

$$\begin{aligned}\pi(p|data) &\propto \left(\sum_{i=1}^M \left\{ \frac{m_i^2 n_i (1-p)^{m_i-2}}{[1-(1-p)^{m_i}]} \right\} \right)^{\frac{1}{2}} \prod_{i=1}^M \left\{ [1-(1-p)^{m_i}]^{x_i} [(1-p)^{m_i}]^{n_i-x_i} \right\} \\ &= [(1-p)^{m_1}]^{n_1-x_1} \prod_{i=2}^M \left\{ [1-(1-p)^{m_i}]^{x_i} [(1-p)^{m_i}]^{n_i-x_i} \right\} \\ &\quad \times \left[\sum_{i=1}^M \left\{ \frac{[1-(1-p)^{m_1}]^{2x_1} m_i^2 n_i (1-p)^{m_i-2}}{[1-(1-p)^{m_i}]} \right\} \right]^{\frac{1}{2}}.\end{aligned}$$

The first term ($i = 1$) of the sum in the square brackets has the form $[1-(1-p)^{m_1}]^{2x_1-1} m_1^2 n_1 (1-p)^{m_1-2}$ and

$$\lim_{p \rightarrow 0^+} [1-(1-p)^{m_1}]^{2x_1-1} m_1^2 n_1 (1-p)^{m_1-2} = 0.$$

The other term ($i > 1$) has the form

$$\frac{[1-(1-p)^{m_1}]^{2x_1} m_i^2 n_i (1-p)^{m_i-2}}{[1-(1-p)^{m_i}]}$$

and

$$\lim_{p \rightarrow 0^+} \frac{[1-(1-p)^{m_1}]^{2x_1} m_i^2 n_i (1-p)^{m_i-2}}{[1-(1-p)^{m_i}]} = 0.$$

Thus, $\pi(p|data) \rightarrow 0$ as $p \rightarrow 0^+$. We can therefore conclude that $\pi(p|data)$ is limited to $(0, 1]$ and therefore $\int_0^1 \pi(p|data) dp$ exists.

Let all $x_i = 0$. Consider two functions, $g(p)$ and $h(p)$, and let

$$g(p) = \frac{1}{\sqrt{p}} \quad \text{for } 0 < p \leq 1$$

and

$$\begin{aligned}h(p) &= [(1-p)^{m_1}]^{n_1-x_1} \prod_{i=2}^M \left\{ [1-(1-p)^{m_i}]^{x_i} [(1-p)^{m_i}]^{n_i-x_i} \right\} \\ &\quad \times \left[\sum_{i=1}^M \left\{ \frac{[1-(1-p)^{m_1}]^{2x_1} m_i^2 n_i (1-p)^{m_i-2}}{[1-(1-p)^{m_i}]} \right\} \right]^{\frac{1}{2}}.\end{aligned}$$

Then

$$\begin{aligned}\frac{h(p)}{g(p)} &= \sqrt{p} h(p) \\ &= \prod_{i=1}^M (1-p)^{m_i n_i} \times \left[\sum_{i=1}^M \left\{ \frac{m_i^2 n_i p (1-p)^{m_i-2}}{[1-(1-p)^{m_i}]} \right\} \right]^{\frac{1}{2}}.\end{aligned}$$

For each $i = 1, 2, \dots, M$ we have,

$$\begin{aligned} \lim_{p \rightarrow 0^+} \frac{p(1-p)^{m_i-2}}{1-(1-p)^{m_i}} &= \lim_{p \rightarrow 0^+} \frac{(1-p)^{m_i-2} - p(m_i-2)(1-p)^{m_i-3}}{m_i(1-p)^{m_i-1}} \\ &= \frac{1}{m_i}. \end{aligned}$$

Thus

$$\lim_{p \rightarrow 0^+} \frac{h(p)}{g(p)} = \sqrt{\sum_{i=1}^M m_i n_i} > 0.$$

Since $\int_0^1 g(p) dp$ converges, $\int_0^1 h(p) dp$ will also converge and we can conclude that $\int_0^1 \pi(p|data) dp$ exists and converges. The posterior distribution is therefore proper and a well-defined distribution. $\square$

3. Simulation Studies

3.1. Simulation Study I - Single Proportion

In this section we will consider a simulation study for a single proportion from pooled samples. A single proportion will be considered where $M = 1$, $M = 2$, $M = 3$ and $M = 4$. We will look at coverage, left noncoverage, right noncoverage, symmetry and interval length. Biggerstaff (2008) defines noncoverage symmetry as the difference in proportional noncoverage, i.e.

$$\text{Symmetry} = \frac{P[\text{Left noncoverage}] - P[\text{Right noncoverage}]}{P[\text{Left noncoverage}] + P[\text{Right noncoverage}]}$$

with a negative value indicating mostly right noncoverage and a positive value indicating mostly left noncoverage. A value of zero indicates symmetric noncoverage.

We considered the different pool size combinations used by Biggerstaff (2008), given in Table 1.

Table 1: Different pool combinations used for the simulation studies in the case of a single proportion.

	Pool size, m	Number of pools, n
$M = 1$	5	200
$M = 1$	50	20
$M = 2$	5	100
	10	50
$M = 3$	10	20
	25	8
	50	12
$M = 4$	5	20
	10	40
	25	12
	50	4
$N = 1000$		
$p = \{0.001, 0.0015, 0.002, 0.005, 0.01\}$		

Table 2 gives the results from Biggerstaff (2008) and the results obtained by us using the Bayesian method. For the Bayesian procedure the highest posterior density (HPD) as well as the equal-tailed intervals are given. The first five intervals in Table 2 are from Biggerstaff (2008). The results in Table 2 are averages taken over the different values for p and the different pool size combinations as given in Table 1.

Table 2: Overall averages of coverage rates, noncoverages, symmetry and average lengths (number per 1 000). Nominal coverage is 95%.

Interval	Coverage	Left non-coverage	Right non-coverage	Symmetry	Length
MIR	0.8070	0.0010	0.1920	-0.99	6.0000
Wald	0.8140	0.0027	0.1830	-0.97	6.5000
Likelihood ratio (LRT)	0.9660	0.0188	0.0150	0.11	7.6000
Profile score	0.9480	0.0476	0.0040	0.84	8.0000
Skewness corrected score	0.9660	0.0205	0.0136	0.20	7.8000
Bayesian (HPD)	0.9584	0.0158	0.0258	-0.34	7.0659
Bayesian (Equal-tail)	0.9375	0.0216	0.0409	-0.20	7.3969

From Table 2 it is clear that the coverage rates obtained by the MIR and Wald intervals are far below the nominal level of 0.95, this was also stated by Biggerstaff (2008). The other five intervals give coverage close to the nominal level. It seems that the HPD credibility interval is somewhat better than the other intervals, because its coverage probability is close to 0.95 with an average interval length that is shorter than those of the other competitors. The coverage rate of the equal-tailed credibility interval is slightly below the nominal level of 0.95. This is because the average coverage probability for $p = 0.002$ is 0.8496. The reason for the low coverage is that $p = 0.002$ falls just outside the credibility interval when $x = 0$, which has a relatively high probability. The symmetry of the two Bayesian intervals are quite satisfactory.

3.2. Simulation Study II - Two Proportions

In this section a simulation study will be considered to obtain confidence intervals for the difference between two proportions. Since the Bayesian procedure is very time consuming only four cases $M_1 = M_2 = 1$, $M_1 = M_2 = 2$, $M_1 = M_2 = 3$ and $M_1 = M_2 = 4$ will be considered. The different combinations of pool sizes are listed in Table 3. In Table 5 the overall averages of the coverage probabilities, left noncoverage, right noncoverage, noncoverage symmetry and mean length for the different parameter values are given. Left and right noncoverage are also defined as distal and mesial noncoverage, respectively.

Steps used for the simulation study for the difference between two proportions: $M_1 = M_2 = 1$

We use simulation to determine the properties of the posterior distribution of the difference according to the following steps, for given values of p_1 and p_2 , and for all possible values of x_1 and x_2 :

- **Step 1**

Calculate the probabilities of outcomes x_1 and x_2 using the binomial distribution, and thus $P(x_1, x_2) = P(x_1)P(x_2)$.

- **Step 2**

Simulate a sample of 100 000 from each of the two marginal posteriors of p_1 and p_2 (from the beta distribution) using Equation 5.

- **Step 3**

Now construct a sample of 100 000 differences, $p_1 - p_2$, and sort them.

- **Step 4**

Stepwise search the sorted sample for the shortest interval containing 95% of the observations, and record the interval, length and mean of the sample.

- **Step 5**

This is now available for every combination of x_1 and x_2 , as well as the probability. So for the given values of p_1 and p_2 , find all the intervals that cover the true value of $p_1 - p_2$ and sum all the corresponding probabilities. This will give the coverage probability. In the same way we find the distal and mesial probabilities and the average length.

Table 3: Different pool combinations used for the simulation studies in the case of the difference between two proportions.

	Pool size, m	Number of pools, n	
$M = 1$	50	20	
$M = 2$	5	100	
	10	50	
	10	50	
	25	20	
$M = 3$	5	100	
	10	40	
	25	4	
	10	50	
	25	12	
	50	4	
$M = 4$	5	20	10
	10	40	20
	25	12	14
	50	4	8
$N = 1\,000$			
$p = \{0.001, 0.0015, 0.002, 0.005, 0.01\}$			

Steps used for the simulation study for the difference between two proportions: $M_1 = M_2 = 2$

The problem here is more complex and there are simply too many combinations of outcomes when M is larger than one. Also we cannot use the beta distribution to simulate from the marginal posteriors of the values of p .

The following steps were used:

- **Step 1**

For a specific data set, say $\underline{x} = [x_{11} \ x_{12} \ x_{21} \ x_{22}]$, we know the form of the marginal posteriors of p_1 and p_2 as given in Equation 6, so we discretise them by calculating their values at small

intervals (0.0001) and then normalise them. Where Equation 6 is given for $M = 2$, where $m_1 = 5, m_2 = 10, n_1 = 100$ and $n_2 = 50$, that is 100 pools of size 5 and 50 pools of size 10.

$$\pi(p|data) \propto \left(\frac{2500(1-p)^3}{[1-(1-p)^5]} + \frac{5000(1-p)^8}{[1-(1-p)^{10}]} \right)^{\frac{1}{2}} \times [1-(1-p)^5]^{x_1} [1-(1-p)^{10}]^{x_2} (1-p)^{1000-5x_1-10x_2}. \quad (6)$$

- **Step 2**

Now we have a probability for every discrete outcome of p_1 and p_2 . Forming all possible combinations of p_1 and p_2 and their associated probabilities by using a grid, we have a distribution for $p_1 - p_2$ for the given $\underline{x}$.

- **Step 3**

After sorting, we can now search for the shortest 95% interval for $p_1 - p_2$, using the associated probabilities and also calculate the mean.

- **Step 4**

Steps 1 to 3 should be done for all possible values of $\underline{x}$. The probability of $\underline{x}$ is the product of the individual binomial probabilities for given p_1 and p_2 .

- **Step 5**

For the given p_1 and p_2 , we find all the values of $\underline{x}$ which yielded an interval that covers the true value of $p_1 - p_2$, and sum their probabilities. This will give the coverage probability. In the same way we find the distal and mesial probabilities and the average length.

Table 4: Averages over all differences for a specific pool size (number per 1 000).

M_i	Coverage	Left noncoverage	Right noncoverage	Symmetry	Length
Equal tailed Intervals					
1	0.9484	0.0316	0.0200	0.2201	12.3060
2	0.9478	0.0324	0.0199	0.2344	11.3186
3	0.9434	0.0355	0.0211	0.2529	11.3626
4	0.9426	0.0358	0.0215	0.2472	11.6218
HPD Intervals					
1	0.9605	0.0285	0.0110	0.4155	12.1883
2	0.9560	0.0302	0.0138	0.3906	11.1702
3	0.9490	0.0352	0.0158	0.3787	11.1586
4	0.9478	0.0355	0.0168	0.3611	11.4601

Steps for the simulation studies of the difference between two proportions in the cases $M_1 = M_2 = 3$ and $M_1 = M_2 = 4$ are similar to those explained above, but more time consuming. Biggerstaff (2008) used a number of models for calculating confidence intervals for the difference between two proportions. Since the Bayesian procedure is very time consuming, only the models given in Table 3, i.e. $M_1 = M_2 = 1$, $M_1 = M_2 = 2$, $M_1 = M_2 = 3$ and $M_1 = M_2 = 4$ will be used for comparison

Table 5: Overall averages of coverage rates, noncoverages, symmetry and average lengths (number per 1 000) for $p_1 - p_2$. Nominal coverage is 95%.

Interval	Coverage	Left non-coverage	Right non-coverage	Symmetry	Length
MIR	0.9320	0.0580	0.0097	0.7100	9.8000
Wald	0.9340	0.0518	0.0139	0.5800	10.6000
Square-and-add Walter	0.9730	0.0126	0.0149	-0.0800	12.9000
Likelihood ratio (LRT)	0.9370	0.0269	0.0358	-0.1400	11.7000
Profile score	0.9630	0.0126	0.0245	-0.3200	15.4000
Skewness corrected score	0.9640	0.0146	0.0217	-0.1900	15.1000
Bias Skewness corrected score	0.9640	0.0146	0.0217	-0.1900	15.1000
Bayesian (HPD)	0.9534	0.0323	0.0143	0.3865	11.4808
Bayesian (Equal-tailed)	0.9455	0.0338	0.0207	0.2387	11.6522

purposes. Inspection of Table 4 shows that adding more models will not have a large influence on the results.

A possibility to speed up the Bayesian simulation procedure is to combine the Bayesian method with the MOVER (Method of Variance Estimates Recovery) which is discussed in Biggerstaff (2008) as the Square-and-Add Walter method.

Table 5 gives the results from Biggerstaff (2008) and the results obtained by us using the Bayesian method. The first seven confidence intervals in Table 5 are from Biggerstaff (2008).

The coverage rate for the HPD credibility interval is above the nominal level of 0.95. This is also the case for the Profile score, Skewness corrected score and Bias Skewness corrected score intervals. For intervals with coverage rates above the nominal level, the HPD procedure yields the shortest interval. The equal-tailed credibility interval also gives good coverage with an interval length that is somewhat shorter than the interval lengths of the other methods. The equal-tailed credibility intervals are more symmetrical than the HPD credibility intervals (this was to be expected) but is not as symmetrical as the Skewness corrected score and Bias Skewness corrected score intervals. The amount of asymmetry in the noncoverage of the HPD credibility interval is not only because of the skewed prior, but also in the method the HPD credibility interval is constructed.

4. Example - West Nile Virus

Biggerstaff (2008) considered an example where a comparison is made between West Nile virus (WNV) infection prevalences in field collected *Culex nigripalpus* mosquitoes trapped at different heights. Biggerstaff (2008) derived asymptotic confidence intervals for the difference between two proportions estimated from pooled samples, where the sizes of the pools are not equal. He considered seven confidence interval methods: an interval based on the Minimum infection rate (MIR), the Wald interval, the Profile score interval, the Skewness corrected score interval, the Bias- and skewness-corrected score interval, Square-and-add Walter (SAW) interval and the Profile likelihood interval. Table 6 summarises the data from Biggerstaff (2008).

For this example the Jeffreys prior will be a suitable prior, since the parameter p is small. By using Jeffreys prior, 95% HPD and equal-tailed credibility intervals are constructed. The results are

Table 6: Summary of *Culex nigripalpus* mosquitoes trapped at different heights of 6m and 1.5m.

	Sample 1 height = 6m	Sample 2 height = 1.5m
Total	2 021	1 324
Number of pools	53	31
Average pool size	38.1321	42.7097
Minimum pool size	1	5
Maximum pool size	50	100
Number of positive pools	7	1

shown in Table 7. Figure 2 shows the posterior distributions (defined in Equation 5), for the 6m and the 1.5m heights.

Table 7: 95% intervals and interval lengths for the difference in proportions (number per 1 000) of the two samples.

		HPD Interval	Equal-tailed Interval	Biggerstaff Interval
Sample 1 height = 6m	95% Interval Length	1.625 - 6.455 4.830	1.666 - 6.500 4.834	1.653 - 7.408 5.755
Sample 2 height = 1.5m	95% Interval Length	0.019 - 3.017 2.998	0.081 - 3.539 3.458	0.044 - 3.670 3.626

From Table 7 it is clear that both Bayesian intervals are shorter than the one obtained by Biggerstaff (2008).

For the mosquito data the posterior distribution of the difference between two proportions is obtained by drawing 100 000 random samples from each of the two posteriors displayed in Figure 2 and each time the difference between the two proportions is calculated. The histogram of the 100 000 differences is illustrated in Figure 3. A normal approximation is also fitted to the simulated differences. From Figure 3 it is also clear that the posterior distribution of the difference between two proportions is somewhat skew.

The 95% credibility intervals as well as the seven confidence intervals calculated by Biggerstaff (2008) are shown in Table 8.

Table 8: Average 95% intervals and interval lengths for the difference between the two proportions (number per 1 000).

	95% Interval	Length
MIR	(-0.250, 5.667)	5.920
Wald	(-0.165, 6.182)	6.347
Profile score	(-0.746, 6.935)	7.681
Skewness corrected score	(-0.572, 6.824)	7.396
Bias- and skewness-corrected score	(-0.570, 6.825)	7.395
Profile likelihood	(-0.355, 6.729)	7.084
Square-and-add Walter	(-0.861, 6.852)	7.713
Bayesian (HPD)	(-0.389, 6.608)	6.997
Bayesian (Equal-tailed)	(-0.433, 6.567)	7.000

The credibility intervals compare relatively well with the other intervals. All the intervals include zero which means that there is not enough evidence to conclude that there is a real difference between

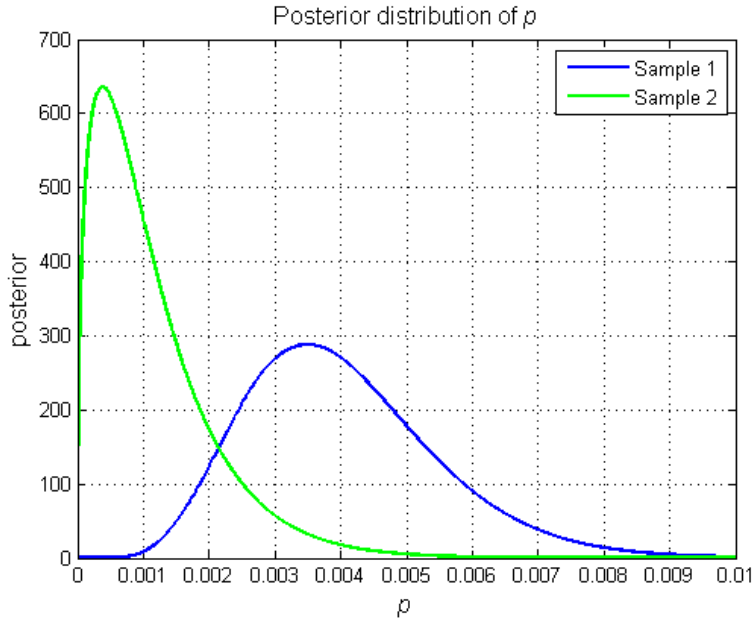


Figure 2: Posterior distribution of p .

the two trapping heights. Although the MIR and Wald intervals are relatively short, they are known for giving poor coverage. The credibility intervals have shorter lengths than the other intervals and do not differ much from the Profile likelihood confidence interval.

5. Conclusion

In this paper two Bayesian confidence methods are compared to results obtained by Biggerstaff (2008). The Jeffreys prior is used for the Bayesian methods since it has some very good properties. It is for example invariant under one-to-one reparameterisation of parameters and in the absence of nuisance parameters it is a first-order probability matching prior, which means that the posterior confidence intervals (although Bayesianly constructed) have good frequentist coverage. Simulation studies are considered as well as an example. The Bayesian methods (HPD and equal-tailed credibility intervals) compare well with the other results and give much better coverage than the Wald and Minimum infection rate (MIR) intervals. The coverage rates of the MIR and Wald intervals are far below the nominal level of 0.95.

For a single proportion the HPD credibility interval seems to be somewhat better than the other intervals because its coverage probability is close to 0.95 with an average interval length that is shorter than those of the other methods. The coverage rate of the equal-tailed credibility interval is slightly below the nominal level of 0.95. This is because the average coverage probability for $p = 0.002$ is 0.8496. As mentioned the reason for the low coverage is that $p = 0.002$ falls just outside

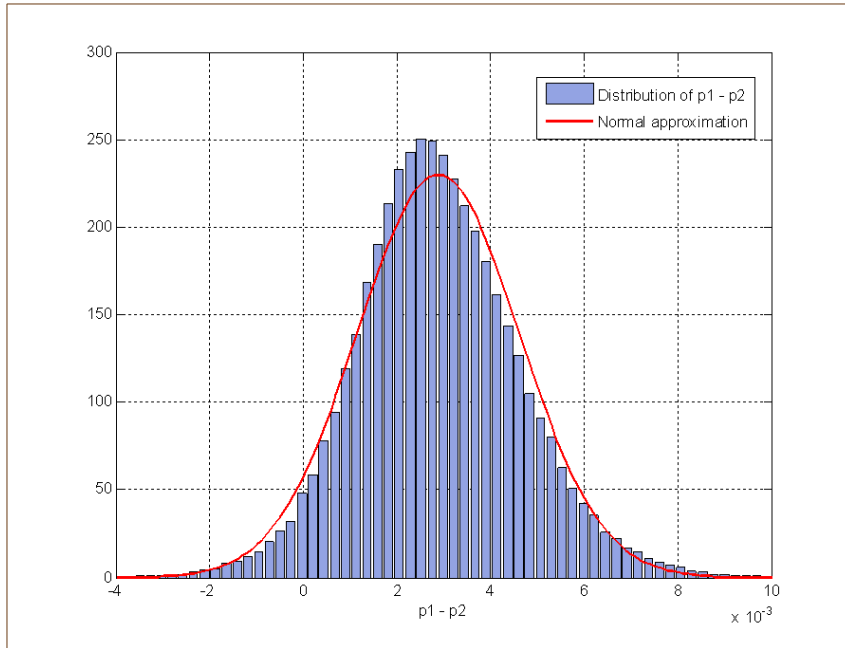


Figure 3: Histogram of $p_1 - p_2$.

the credibility interval when $x = 0$, which has a relatively high probability. Brown et al. (2001) mentioned that this erratic behaviour of the coverage probability of the binomial confidence intervals can get downright extreme in certain practically important problems. They however suggested the equal-tailed Jeffreys prior interval and an interval derived by Agresti and Coull (1998) for larger n . A frequentist justification for use of the Jeffreys interval was also provided by them. The symmetry of the two Bayesian intervals are also quite good.

Inspection of Table 5 shows that the coverage rate for the HPD credibility interval is above the nominal level of 0.95. This is also the case for the Profile score, Skewness corrected score and Bias Skewness corrected score intervals. For intervals with coverage rates above the nominal level, the HPD interval method yields the shortest interval. The equal-tailed credibility interval also gives good coverage with an interval length that is somewhat shorter than the interval lengths of the other methods. The equal-tailed credibility intervals are more symmetrical than the HPD credibility intervals (this was to be expected) but it is not as symmetrical as the Skewness corrected score and Bias Skewness corrected score intervals.

For the mosquito data the credibility intervals have shorter lengths than the other five intervals and do not differ much from the Profile likelihood confidence interval. It is further clear from the example and simulation studies that the posterior distribution for the difference between two proportions is more symmetrical than the posterior distribution for individual proportions.

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